# Escape the Streets Traffic Education Game

Team Geralt
Master of Applied IT
Fontys University of Applied Sciences
Eindhoven

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## Abstract

Escape the Streets is a serious game that lets players practise decisions in traffic while delivering packages. This study examines whether participants answer more traffic questions correctly after playing and how they experience the game. Participants will complete a questionnaire before and after playing the game. A five-point rating after play will measure self-reported enjoyment. *[Add the number of participants, main findings and conclusion after testing.]*

## 1 Introduction

The starting point for this project was a case about making streets in Eindhoven safer. We focused on traffic education for children from 6-12 years old. We developed Escape the Streets to let players experience traffic situations from three different categories, a pedestrian, cyclist and van driver. Per category there are multiple levels developed.

The player collects a package at point B and has to deliver it to point C within five minutes. During the delivery the player has to take care of crossings, traffic lights and other road users. If you cause a violation it costs a star and triggers a message explaining the mistake. We want to find out whether players understand these situations better after playing and whether they find the game fun. We also wanted to explore engagement.

## 2 Traffic education and baseline

Dutch primary schools decide how much time to spend on traffic education. Children also learn traffic during everyday trips with their parents or carers [1]. Veilig Verkeer Nederland (VVN) combines classroom activities, practical lessons and home exercises. The materials include digital exercises with feedback [2].

SWOV reports that some primary-school traffic education programmes improve knowledge and show small changes in behaviour, while the evidence varies between programmes [1]. The WEVER evaluation also found that its school data could not establish the effects of individual education programmes [3].

One study close to our game evaluated WegWijsVR. After six weeks of additional virtual practice, the percentage of tasks completed safely in the virtual assessment increased from 16% to 49%. The group continuing the usual school lessons showed a change of approximately one percent. The children who

rated the training gave it an average of 8 out of 10 [4]. These findings give us a reference for virtual traffic training. The different tasks and amount of practice limit direct comparison with our playtest.

For our own study, each participant's score on the form before playing is their starting baseline. We compare it with their score after playing.

## 3 Research question and hypotheses

**Research question.** How do participants their scores on traffic questions change after playing *Escape the Streets* and how much do they enjoy playing it?

H1. On average, participants will answer more traffic questions correctly after playing the game compared to before playing the game.

H2. Most participants will report that they enjoyed playing the game.

For H2, we count the fun at a five-point fun scale. With level 5 as most fun experienced.

## 4 Methodology

### 4.1 Design and participants

We used a single group with a form before and after playing. The intended audience of *Escape the Streets* is children aged 6 to 12. For this first test, we recruited a mixed group of students from Fontys. Their responses can help us find problems with the game and explore changes in scores before and after playing. **[Add the number of participants, their age range and the recruitment details.]**

### 4.2 Questionnaire design

We used Microsoft Forms to create a questionnaire. Participants fill in whether they are completing the form before or after playing. The quiz includes six multiple-choice questions. We chose the cyclist perspective to test with the participants. Each question contains of five answer options. The images show the situations from the game per question. Correct answers are hidden from participants. When preparing the traffic questions, we looked at the VVN refresher quiz for examples of how to present traffic situations and answer options [5]. We included questions of different difficulty, with some more challenging questions to leave room for improvement before and after playing the game. If participants already score 100% before playing, their scores cannot improve. This is called a ceiling effect.

We used the same traffic questions before and after play, with the answer options shown in a counterbalanced order. This makes it harder to repeat a choice by remembering where it appeared. Participants can still recognise the question and remember the wording of the answer they chose.

### 4.3 Procedure

Before playing, participants enter their name and answer the traffic questions. They then receive a short explanation of the controls and play levels 3, 4, 6, and 8 of the cyclist's perspective. They get up to five minutes per level. We selected these levels because these traffic situations most closely match those covered by the questionnaire.

After playing participants fill in the same identity as before and answer the same traffic questions and rate how fun the game was. We use their identity to match the two forms. If an email address is available on both submissions, we can use it to check the match.

### 4.4 Measures and analysis

For each participant who completed both forms, we count the correct answers before and after playing. The difference between the two scores shows how much that participant's score changed. We report the number of matched pairs, the average score before and after play, and how many scores went up, stayed the same or went down. We also report how many participants reached the ceiling effect.

We use the five-point rating for the question: How fun was your experience with playing the game? We analyse ratings from submissions marked as after playing and report how many participants chose each option, including the proportion who chose one of the two highest options. We treat this as a measure of enjoyment.

## 5 Results

**[Complete after the playtest. Actual sample, number of matched pairs, scores before and after, individual changes, fun ratings, any observations of engagement recorded during play. Feedback or stuff]**

## 6 Discussion and limitations

We tested with Fontys students, while the game is intended for children aged 6 to 12. The students may have different traffic knowledge, reading skills and experience with games. Their feedback can guide changes to the prototype. Testing with children is needed to assess whether the game and questionnaire are suitable for the intended age group.

We used the VVN quiz as a reference when writing the questions. The resulting questionnaire has not been formally validated. The more difficult questions were included to reduce the chance of a ceiling effect, although some participants may still score full marks before playing.

Changing the order of the answer options reduces reliance on remembering an option's position. Participants may still recognise the questions and recall their previous answers. With one group, it is difficult to separate changes due to playing from changes due to seeing the questions twice. Given the time available, we used the same items for this first test. A follow-up study could use two larger sets of validated questions that cover the same topics at a similar difficulty, with one set before play and the other afterwards.

The fun rating measures participants' enjoyment. The form has no separate question on engagement, such as whether the game held their attention or whether they wanted to keep playing. We can use any observations recorded during play as exploratory evidence of engagement.

*[Discuss what the results show for each hypothesis and which game improvements follow from the feedback.]*

## 7 Conclusion

*[Answer the research question using the observed results. State what the study supports and which game change should be tested next.]*

## References


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[4] G. J. Wijlhuizen and S. T. van der Kint. *Verkeersvaardigheidstraining met virtual reality voor basisschoolleerlingen*. R-2018-12. SWOV, 2018. URL: https://swov.nl/system/files/publication-downloads/r-2018-12.pdf.

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